

Advances in Bridge Management and Inspection

Formerly



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Goals for Inspection and Management

- Ensure optimal safety and operational capability in the most efficient manner.
 - Inspection is used as the eyes and ears of the program to find and document the current condition including any problems
 - Management utilizes the inspection data along with the organization's priorities to determine the most efficient way to ensure goals are met (i.e. safety, performance, capacity)



Technology + Bridge Industry

- Technology availability is different from industry adoption

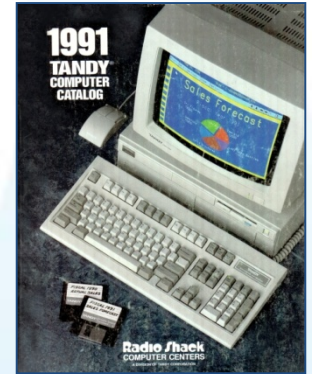


Bridge Industry:

- Safety Focused
 - Utilize only proven approaches
 - Code/rule based
- Government agencies are primary entities
- Skilled Professionals
- Changes are often slow to be implemented

Past ... Present ... Future

- Remember technology 20 years ago
- Technology today
- Imagine technology 20 years from now



AASHTOWare Bridge Management

- Utilized by 44 DOTs plus local and intl. agencies
- BrM/Pontis 5.2 is funded by a voluntary participation from more than 20 state DOTs, under DOT guidance and expertise
- Tools that are easier to use and understand:
 - Planning
 - Deterioration
 - Risk
 - Multi-objective analysis
 - Lifecycle costs
 - Project models
 - Dashboards
 - Corridor planning



AASHTOWare Bridge Management

www.AASHTOWareBridge.com



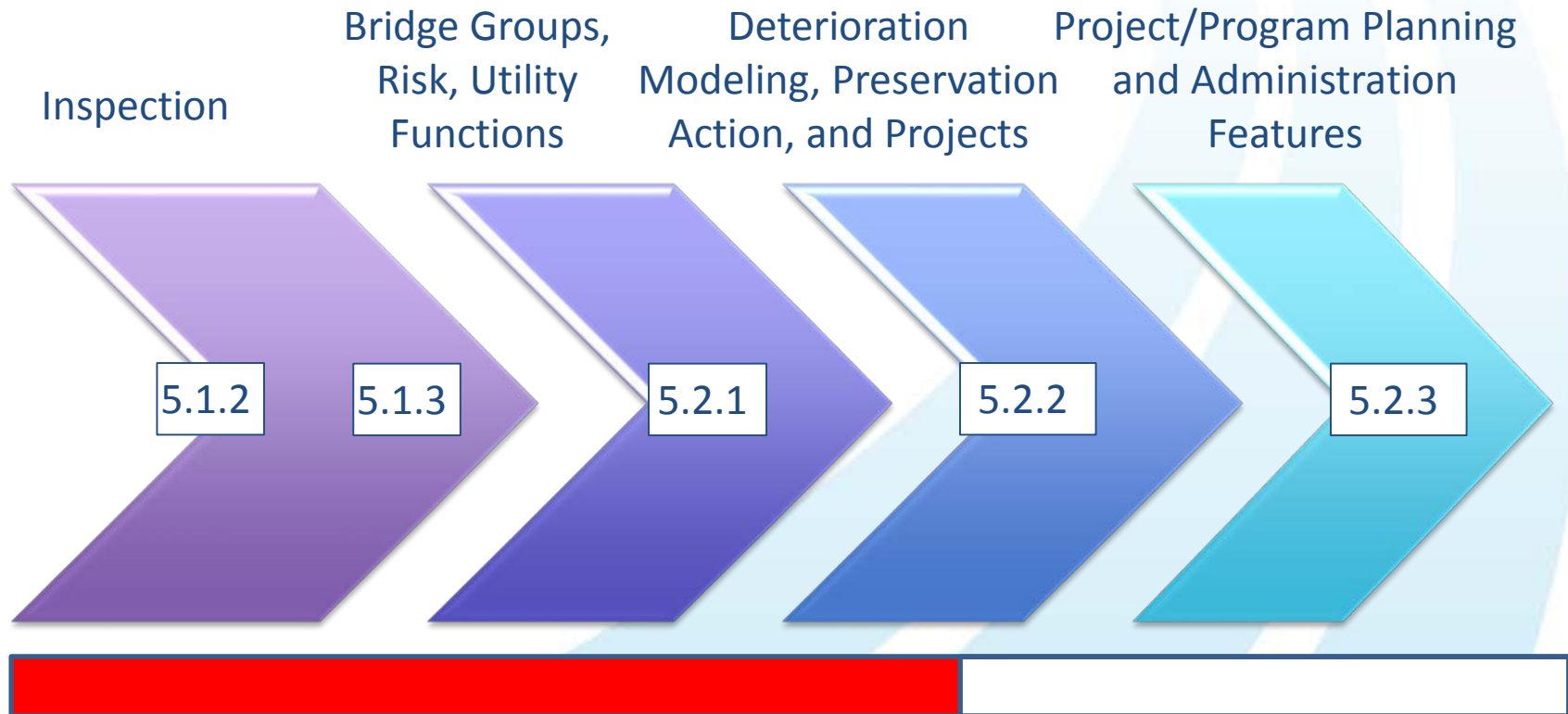
Basic Approach of 5.2

- Utilize extensive research and lessons learned over past 20 years
 - Continue to evaluate best approach and layout
 - Interactively working with Task Force
 - TRT Expert Panel of State Representatives
- Multiple Phased implementation
- Integrate full AASHTO and FHWA requirements

Bridge Management 5.2 Stages

- Development on 5.2 is rapidly moving forward with coordinated efforts between the Task Force, TRT, and Contractor
- Phased releases
 - Version 5.1.2/5.1.3 (Mar 2012 / May 2013)
 - New inspection and inventory functionality, integration with mapping
 - Version 5.2.1 (Feb 2014)
 - Core program framework, risk assessments, integrated utility functions, network corridors
 - Version 5.2.2 (Mid-2015)
 - Implementation of new deterioration models and multi-objective analysis
 - Version 5.2.3 (Planned 2016)
 - Integrated project and program planning
 - All administrative features

Bridge Management 5.2 Stages



Bridge Management 5.2.1

- Released in Early 2014
- Key Features
- Google Mapping Functions
- Utility Functions
- Needs Prioritization
- Support For 2013 Element Inspection Manual
- Cross-Browser Support
- Key User Requests



Key Parts of Pontis/BrM 5.2

- What do I have?
- What Condition is it in?

**Bridge
Inventory**

**Bridge
Conditions**

Key Parts of Pontis/BrM 5.2

- What Risks Do I have?

Bridge
Inventory

Bridge
Conditions

Risk

Key Parts of Pontis/BrM 5.2

- What are my identified needs?

**Bridge
Inventory**

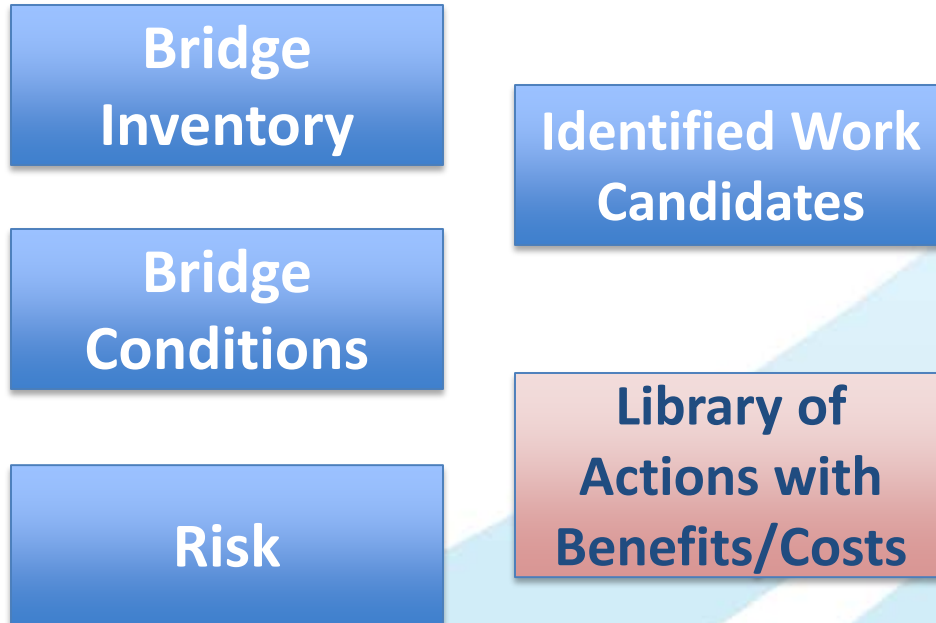
**Bridge
Conditions**

Risk

**Identified Work
Candidates**

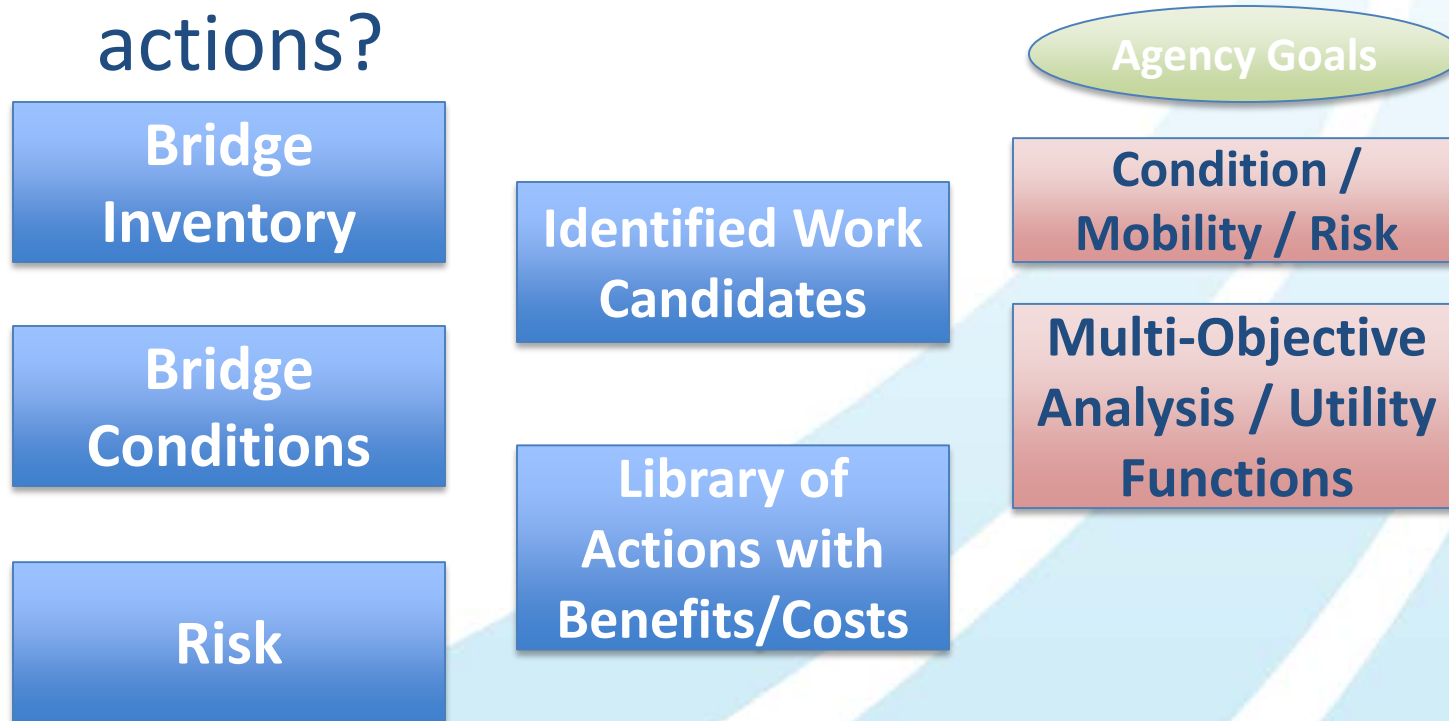
Key Parts of Pontis/BrM 5.2

- What are the benefits of the work candidates?



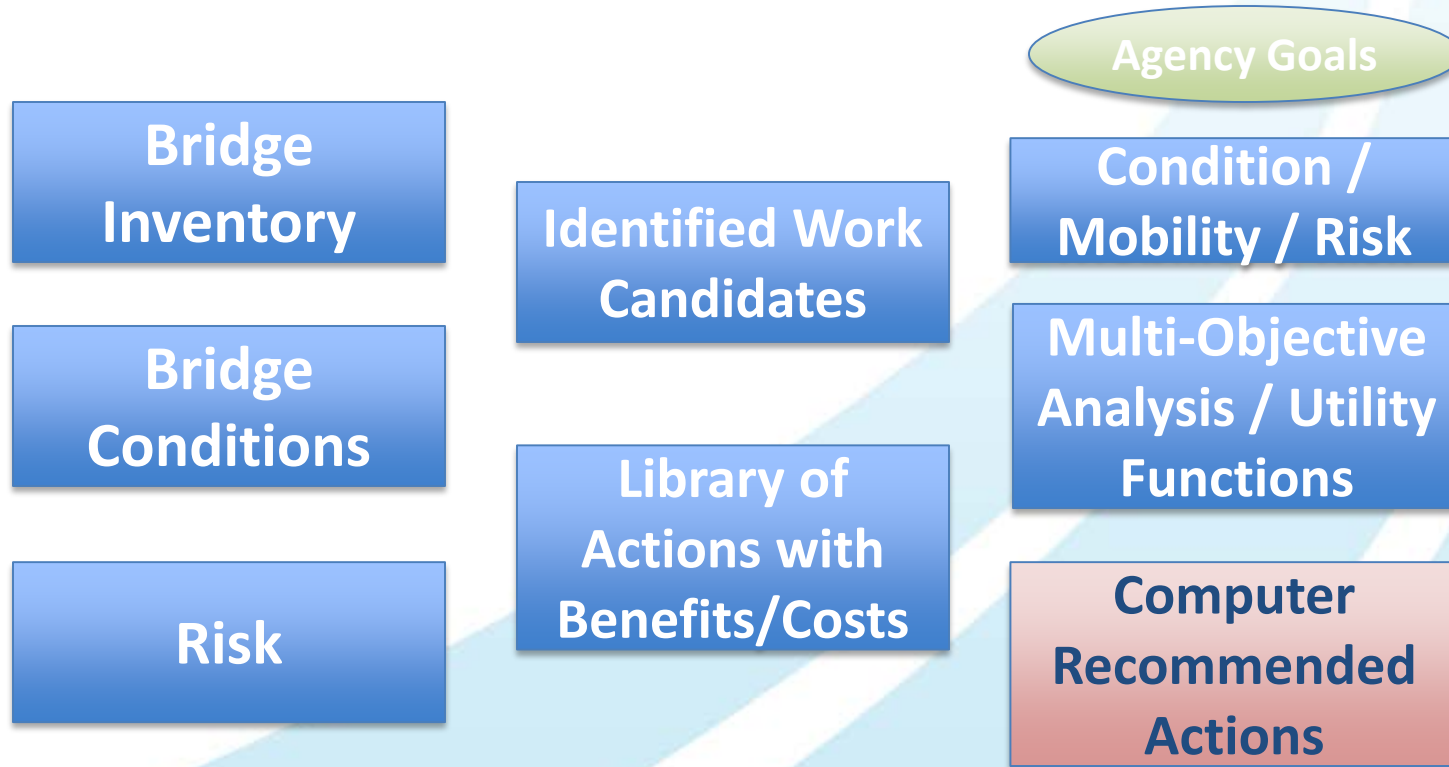
Key Parts of Pontis/BrM 5.2

- How do I compare benefits of very different actions?



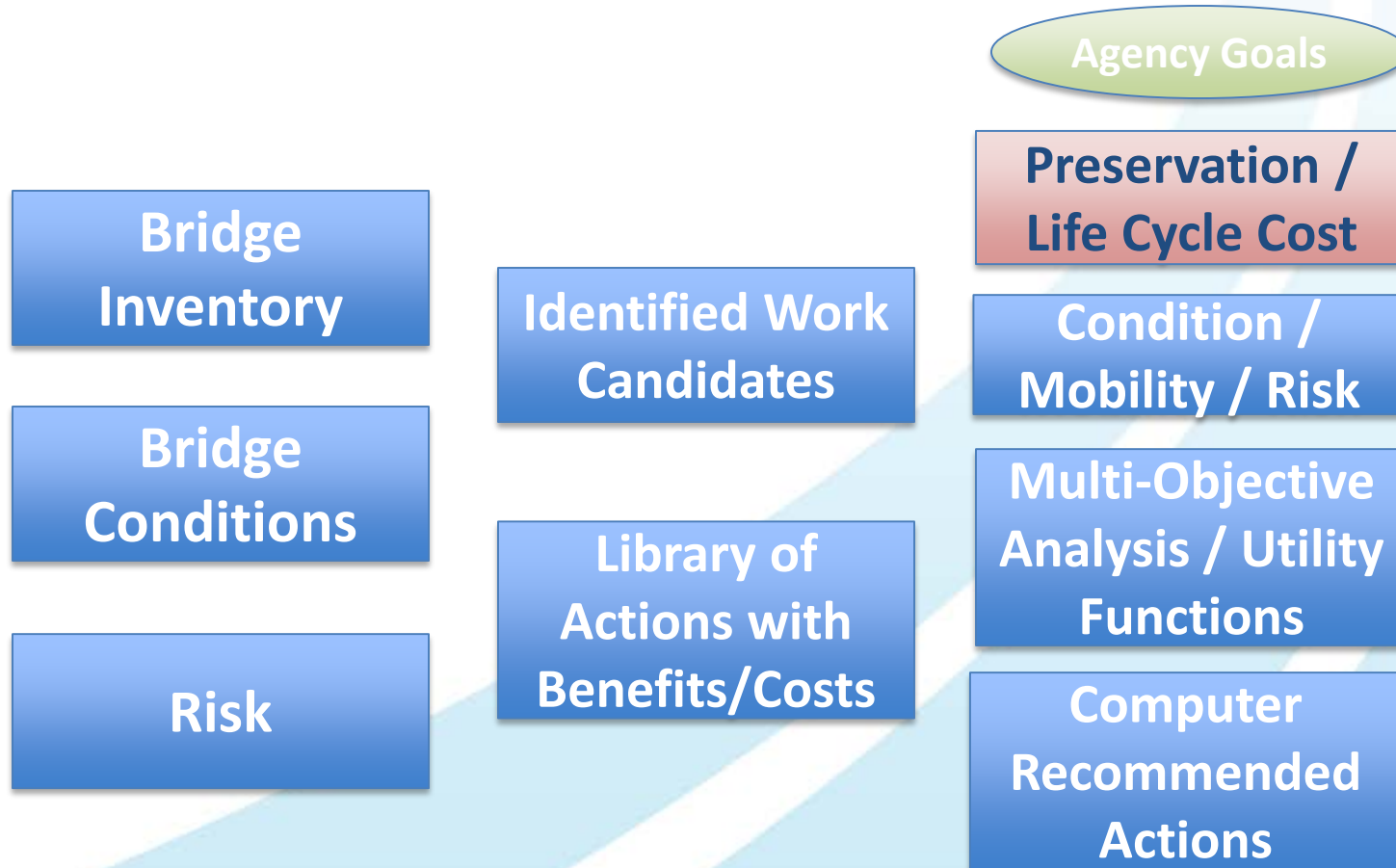
Key Parts of Pontis/BrM 5.2

- What actions should I possibly take?



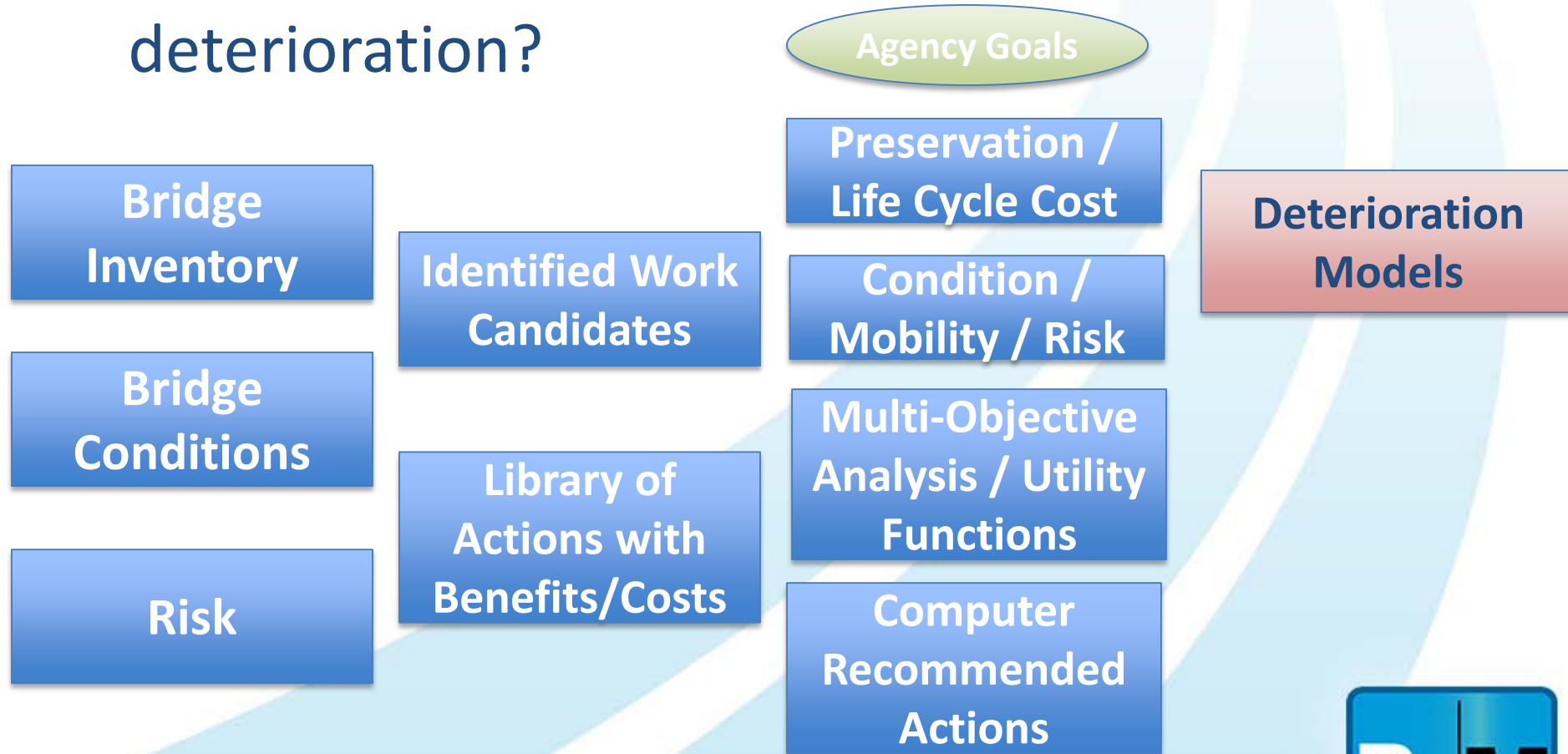
Key Parts of Pontis/BrM 5.2

- What are the benefits of preservation actions?



Key Parts of Pontis/BrM 5.2

- What is the effect of future time and deterioration?



- Grouping/comparing needs across bridges into projects and programs



Improved Decision Making Tools



Better fit for agency workflow and business processes

AASHTOWare Bridge Management

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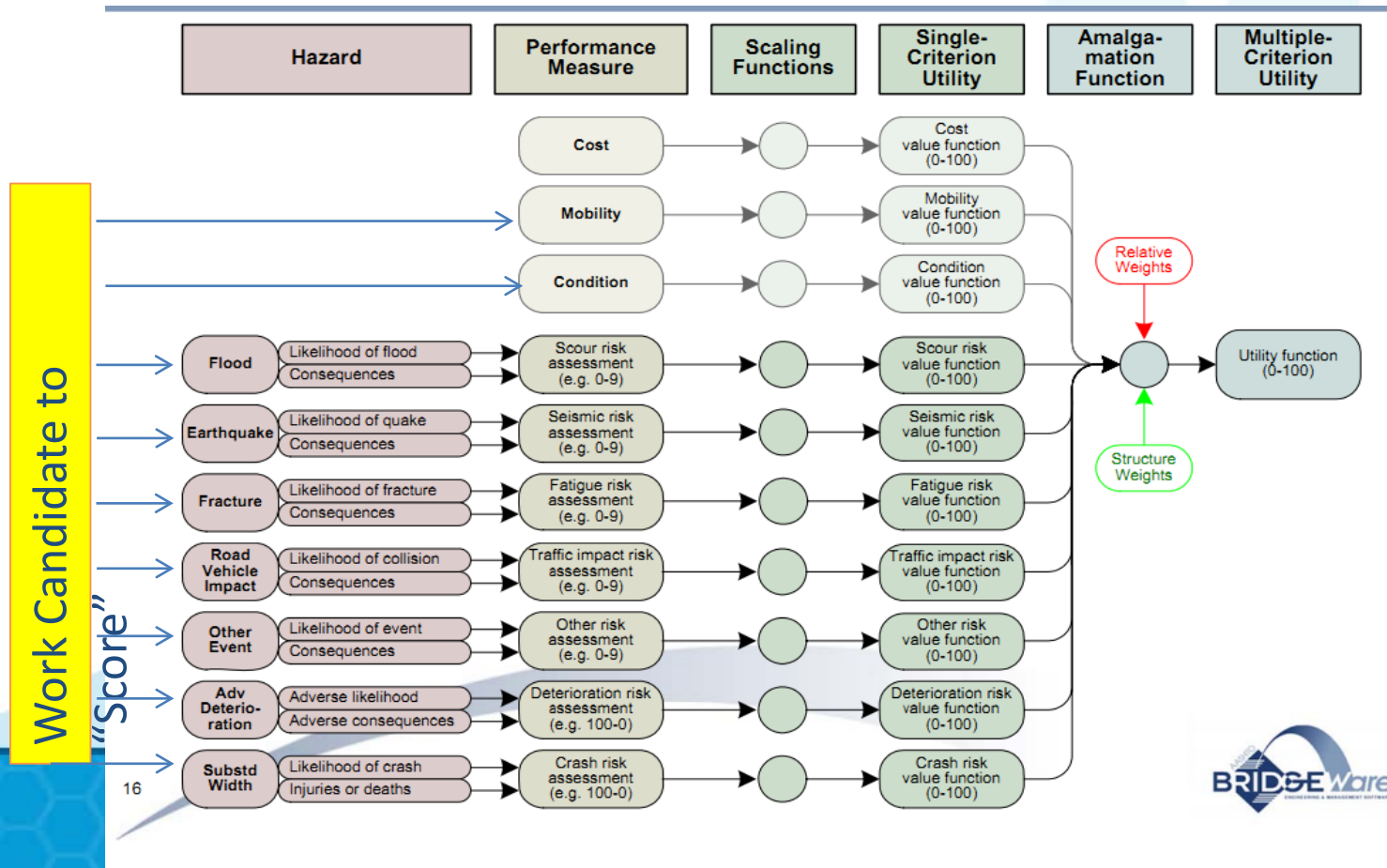


Implementing Utility Functions

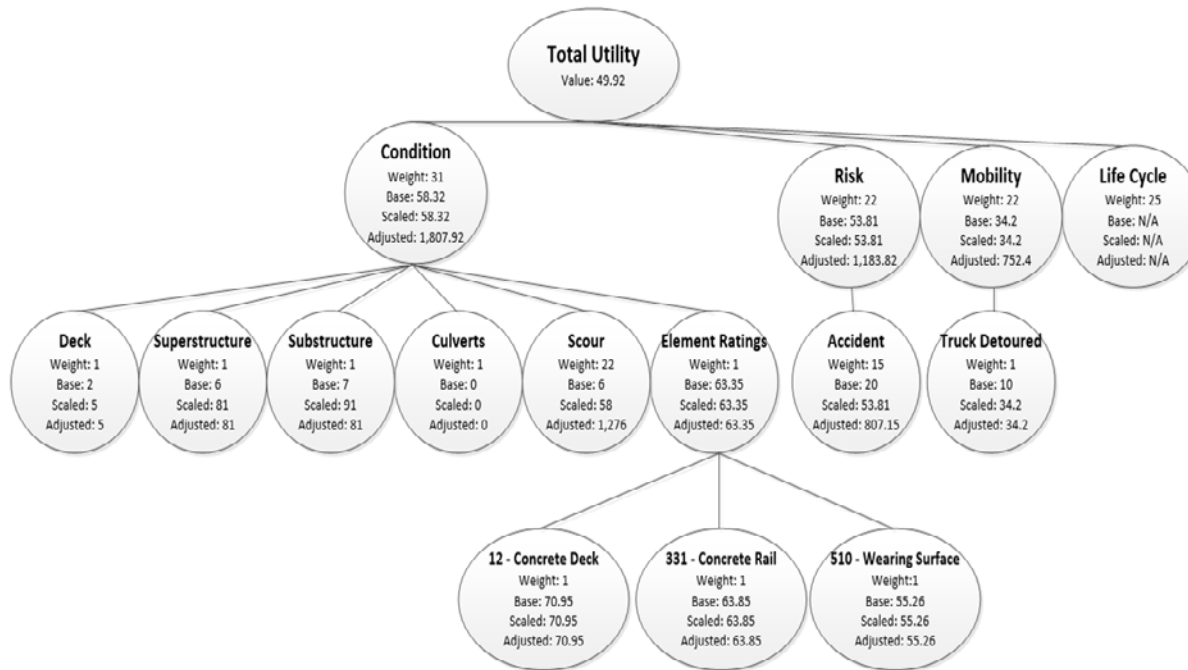
- Multi-objective framework that can be used to show the value (utility) of an action for a bridge
- Utility will also be shown for each sub-area
 - Mobility
 - Condition
 - Risk
 - Life Cycle Cost (deterioration models + preservation benefits)
- Work candidates (preservation and others) are evaluated for how they contribute to mobility, lifecycle cost, condition and risk weightings
- Allow for comparison of current and future status and benefits of actions

Multi-Objective Analysis Framework

- The model will score each work candidate identified.



Multi Objective Analysis



Parent Category Base Value = $\sum \text{Child Adjusted Values} / \sum \text{Child Weights}$
 Scaled Value = Determined based on defined scaling in Admin Utility
 Adjusted Value = Scaled Value * Weight

- In this example, the bridge has a utility value of 49.92
- In this example, the Condition component is weighted the most heavily of all components of a bridge (weighted to be more important than Risk, Mobility, and Life Cycle)
- In this example, an agency is able to see exactly how every component and subcomponent exactly impacts the overall utility of the asset

Multi-Objective/Utility Function



By incorporating utility functions, BrM is able to combine elements of Risk, Lifecycle Cost, Condition, Mobility, and other agency defined criteria to calculate the utility or value of a particular bridge.

Full Transparency for Utility Values



Welcome: **AASHTOWare**
Bridge Management
Database: **BrM_Demo**
[Help ?](#) [Account](#) [LogOut](#)

Menu

Work Candidates

Utility Value

Bridges Reports Admin Inspection Gateway Analysis

Utility Value: 45.98

Condition Value

Base Condition Value: 47.29 Scaled Value: 47.29 x Weight: 31.00 = Adjusted Condition Value: 1465.99

Condition Item	Base Value	Scaled Value	Weight	Adjusted Value
Deck	4	42	10.00	420.00
Superstructure	6	81	1.00	81.00
Substructure	7	91	1.00	91.00
Culverts			1.00	
Element ratings	22.74	22.74	1.00	22.74

Risk Value

Base Risk Value: 55.90 Scaled Value: 55.90 x Weight: 22.00 = Adjusted Risk Value: 1229.80

Risk Item	Base Value	Scaled Value	Weight	Adjusted Value
Scour	6	58	1.00	58.00
Accident	20.00	53.81	1.00	53.81

Mobility Value

Base Mobility Value: 34.2 Scaled Value: 34.2 x Weight: 22.00 = Adjusted Mobility Value: 752.40

Mobility Item	Base Value	Scaled Value	Weight	Adjusted Value
Percent of truck detoured.	10	34.20	1.00	34.20

LifeCycle Value

Base LifeCycle Value: N/A Scaled Value: N/A x Weight: 25.00 = Adjusted LifeCycle Value: N/A

LifeCycle Item	Base Value	Scaled Value	Weight	Adjusted Value
No records to display.				

Preservation (and other) Actions

- Default Actions supplied by State TRT members
- Actions have a default cost
- Attach actions to benefit groups

The screenshot displays the AASHTOWare Bridge Management software interface. The top navigation bar includes tabs for Bridges, Reports, Admin, Inspection, Gateway, and Analysis. A left-hand menu lists various functions such as Security, Parameters, Element Spec, Definitions, Actions, Action Defs, Options, Cost Index, Formulas, Export Options, Data Dict., Checked Out Bridges, Renumbr Brkey, Logging, Navigation & Field Security, Sync Mapping, and Utility. The main content area is titled 'Action Defs' and contains a table of predefined actions. Below this table is a section for 'Associated Benefit Group' with a dropdown menu set to '1 - Deck Rehab'. At the bottom of the interface is a 'Save' button.

Key	Name	Description	Notes	Sort Order
26	Channel-Regrde Channel	Channel-Regrade Channel Under Bri	-1	
27	Channel-Remove Debris	Channel-Remove Debris	-1	
28	Channel-Repair Washout:	Channel-Repair Washouts / Erosion	-1	
29	Deck-Install Protection Sy	Deck-Install Protection System	-1	
30	Dck-Ptch spls->Dck-Repr	Deck-Patch spls->Deck-Repair (Po	-1	
31	Deck-Place Overlay	Deck-Place Overlay	-1	
32	Deck-Rehab	Deck-Rehab	-1	

ID	Element Name	Cost Per Unit	Unit
12	Concrete Deck	\$ 74.32	sq.ft
38	Concrete Slab	\$ 74.32	sq.ft

Benefit Groups

Welcome: AASHTO
Bridge Management
Database: BrM Demo
Help ? Account LogOut

Bridges Reports Admin Inspection Gateway Analysis

Benefit Groups

+ Add new record

Benefit Group Name	Description	Action Defs	Edit	Delete
Deck Rehab	Rehab a concrete deck	Deck-Rehab	Edit	Delete
Bridge rail repair	Repairs done to the bridge rail	Bridge Rail-Rehab, Bridge Rail-Repair	Edit	Delete
Approach Slab repair	Repairs to the approach slab	Approach Slab-Repair	Edit	Delete

Deck Rehab Elements

+ Add new record

Element Name	Origin State	CS1	CS2	CS3	CS4	Edit	Delete
- Element: (38) Concrete Slab							
(38) Concrete Slab	CS3		100%			Edit	Delete
(38) Concrete Slab	CS4		100%			Edit	Delete
- Element: (12) Concrete Deck							
(12) Concrete Deck	CS3		100%			Edit	Delete
(12) Concrete Deck	CS4		100%			Edit	Delete

Deck Rehab Fields

+ Add new record

Table Name	Column Name	New Value	Increment	Edit	Delete
inspevnt	dkrating	8		Edit	Delete

Deck Rehab Risks

+ Add new record

Assessment Name	New Value	Increment	Edit	Delete
Accident		-5	Edit	Delete

Save

- Benefit Groups which define what effects an action has
- Benefit Groups can apply to any elements, fields, and/or risks.

Individual Bridge Analysis Pages show benefits of actions

The screenshot displays the AASHTOWare Bridge Management software interface. The top navigation bar includes links for Bridges, Reports, Admin, Inspection, Gateway, and Analysis. The main content area is titled 'Bridge 04 07598' and 'Name: Western Canal Bridge'. It provides a detailed view of the bridge's characteristics, including its description, mobility and safety, conditions, current scaled performance, and sufficiency. A table at the bottom lists 'Work Candidates Existing for the Selected Bridge' with columns for Work Candidate, Utility, Utility Change, Cost, Benefit / Cost (\$K), and Cost (\$K) / Benefit.

Work Candidate	Utility	Utility Change	Cost	Benefit / Cost (\$K)	Cost (\$K) / Benefit
Do Nothing	45.98				
04 0759-NIMO-091313-E2A8C9DA58 - Approach Slab-Repair	46.38	0.4	\$10,000.00	0.04	\$25
A-DOT001-057F6A47-0000006A - Bridge Rail-Repair	46.67	0.69	\$50,000.00	0.0138	\$72.4638
A-DOT001-057F6A47-0000006D - Deck-Rehab	64.54	18.56	\$100,000.00	0.1856	\$5.3879

- Provides snapshot of work candidates and a detailed view of effect of each
- View includes all related utility value information and criterion

Recommended Actions

- Calculates the cost/benefit ratio for all possible actions
- Ranks all available actions providing recommendations on what should be applied to a bridge

Menu | Bridges | Reports | Admin | Inspection | Gateway | Analysis

Bridge 04 07598 | Name: Western Canal Bridge | Facility Carried (007): JESSE OWENS PKWY | Feature Intersected (006A): WESTERN CANAL

Description

Route:	00000	Milepoint:	mi
District:	District 2	County:	Maricopa
Func:	17 Urban Collector	Area:	02B - Paul Goldsmith
Owner:	City/Municipal Hwy Agenc	Resp:	City/Municipal Hwy Agenc
Material:	5 Prestressed Concrete	Design:	05 Multiple Box Beam
Scour:	6 Calcs not made		

Mobility - Safety

Operating:	1 LF Load Factor	Inventory:	1 LF Load Factor
Geometry:	9 Above Desirable Crit	Approach:	8 Equal Desirable Crit
Waterway:	8 Equal Desirable		

Conditions

Deck:	4 Poor	Superstr:	6 Satisfactory
Substr:	7 Good	Culvert:	N N/A (NBI)
Structure:	8 Protected	Deck Index:	50
Superstr Index:	44.44	Substr Index:	
Culvert Index:		Structure HI:	42.1

Current Scaled Performance

Condition:	47.29	Risk:	55.9
Lifecycle:		Mobility:	34.2

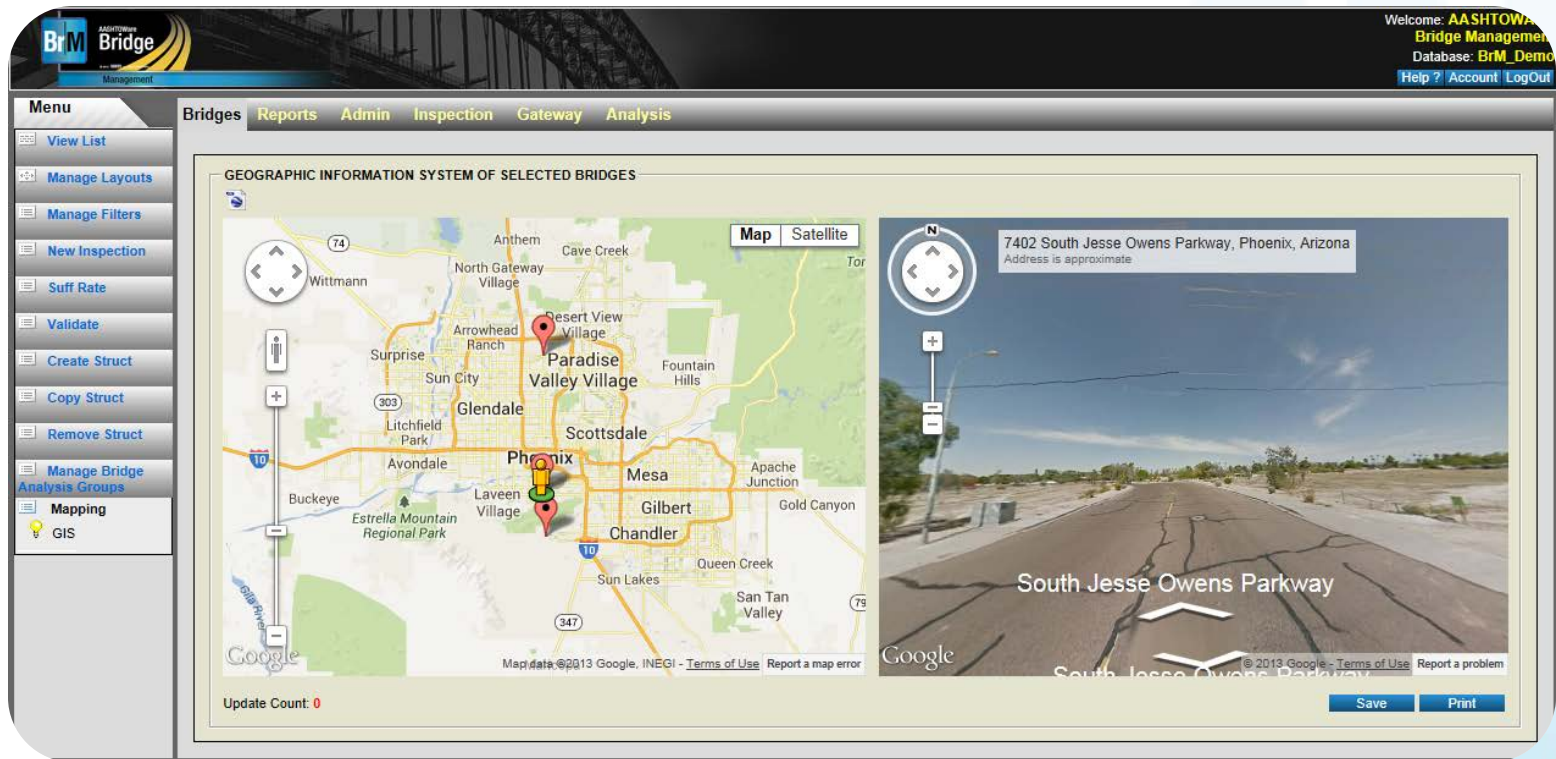
Sufficiency

Rating:	97.5	SD/FO:	Not Deficient
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All Available Work Candidates

Work Candidate	Base Utility	Utility	Utility Change	Estimated Cost	Benefit / Cost (\$k)
Approach Slab-Repair	45.98	46.38	0.4	\$1,114.84	0.3588
Bridge Rail-Rehab	45.98	46.67	0.69	\$45,720.00	0.0151
Bridge Rail-Repair	45.98	46.67	0.69	\$45,720.00	0.0151
Deck-Rehab	45.98	64.54	18.56	\$148,644.86	0.1249

Visualize Needs with Maps



Bridge Analysis Groups to Assist with Preservation

- Ability to create groups based on any inventory criteria combination
- i.e. Steel bridges with ADT over 30,000.
- Apply analysis to groups.

The screenshot displays the 'Bridge Analysis Groups' form within the AASHTOWare Bridge Management software. The interface includes a 'Menu' on the left with options like 'View List', 'Manage Layouts', 'Manage Filters', 'New Inspection', 'Suff. Rate', 'Validate', 'Create Struct', 'Copy Struct', and 'Remove Struct'. The main form area is titled 'ALL BRIDGE ANALYSIS GROUP FILTERS' and includes a 'Filter' dropdown set to 'BrM - None' and a 'Layout' dropdown set to 'Bridge Analysis Group'. Below this is a 'Bridge Analysis Groups Quick Filter' section with a 'User' dropdown and a 'Load filter' button. The form is divided into two main sections: 'Roadway Characteristics' and 'Bridge Characteristics'. The 'Roadway Characteristics' section includes fields for 'Route On Structure' (005A), 'Kind of Highway' (005B), 'Level of Service' (005C), 'Rte' (005D), 'Directional Suffix' (005E), 'Roadway Name', 'Kilometer Post' (011), 'Average Daily Traffic' (029), 'Truck Percentage' (109), 'Number of Lanes' (028A), 'NHS Status' (104), and 'Functional Class' (026). The 'Bridge Characteristics' section includes fields for 'District' (002), 'Admin. Area' (500 - Not NBI), 'County' (003), 'Owner' (022), 'Custodian' (024), 'On/Off State System', and 'Bridge Group'. At the bottom, there is a 'Create' button and a table titled 'Bridge Analysis Groups' with columns for 'Bridge Analysis Groups', 'Bridge_Count', 'Roadway_Count', and 'Description'. The table contains two rows: 'New Bridge_Groups 9172013_235' and 'New Bridge_Groups 9172013_4', both with counts of 0 and a description 'Created by software, please replace with valid description.'.

Bridge Analysis Groups	Bridge_Count	Roadway_Count	Description
New Bridge_Groups 9172013_235	0	0	Created by software, please replace with valid description.
New Bridge_Groups 9172013_4	0	0	Created by software, please replace with valid description.

Deterioration Modeling

- Implement new deterioration model logic
 - Weibull approach to include time factor
- Easy to construct/new elicitation process
- Utilize AASHTO Elements
 - Protective Systems
 - Defect Flags
- Allow for Multi-path deterioration

Deterioration Modelling

- An agency is able to see the direct impact of performing work on an asset, and how it will impact the bridge currently, as well as years into the future
- Also able to see the *direct impact of performing work at a later point in time*. This aids an agency in the decision to determine when the optimized time would be to perform the selected work.

Deterioration Modeling and Multi Objective Analysis (example)

Work Candidates Existing for the Selected Bridge

Work Candidate	Utility	Utility Change	Cost	Benefit / Cost (\$)	Cost (\$)/ Benefit	Action Year
Do Nothing	49.92					0
04 0759-NIMO-041614-27F109E126 - Approach Railing-Repair	50.5	0.58	\$2,000.00	.29	\$3	0

Page size: 10
 2 items in 1 pages

Effects on Each Utility Criterion

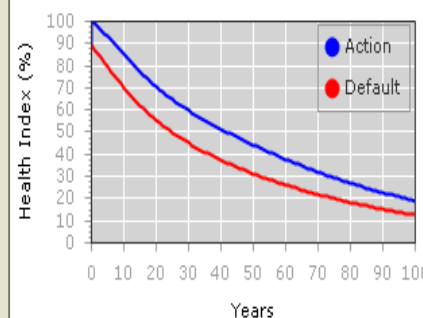
Category name	Before WC	After WC
Total Utility	49.92	50.5
Condition	58.32	59.73
Deck	5	5
Superstructure	81	81
Substructure	91	91
Scour	58	58
Element ratings	63.35	100
(12) Reinforced Concrete Deck	70.95	100
(331) Reinforced Concrete Bridge Railing	63.85	100
(510) Wearing Surfaces	55.26	100
Risk	53.81	53.81
Accident	53.81	53.81
Mobility	34.2	34.2
Percent of truck detoured.	34.2	34.2

Effects on Each Element

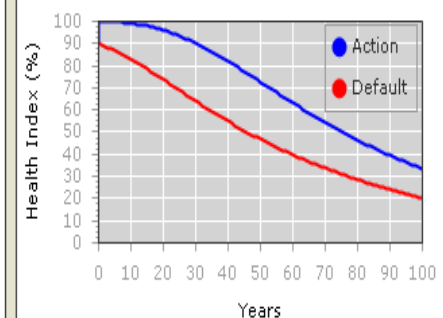
Element	Str. Unit	Env.	Condition	Effect
> (12) Re Concrete Deck	0	Low (2)		
☒ (331) Re Conc Bridge Railing	0	Low (2)		
(510) Wearing Surfaces	0			

Deterioration

★ Bridge



★ Element: 331



Deterioration Modeling and Multi Objective Analysis (example)

Work Candidates Existing for the Selected Bridge

Work Candidate	Utility	Utility Change	Cost	Benefit / Cost (\$k)	Cost (\$k) / Benefit	Action Year
Do Nothing	49.44					0
04 0759-NIMO-041614-27F109E126 - Approach Railing-Repair	50.5	1.06	\$2,000.00	.53	\$2	20

Page size: 10
 2 items in 1 pages

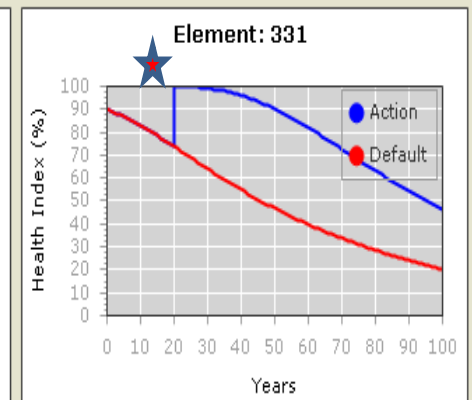
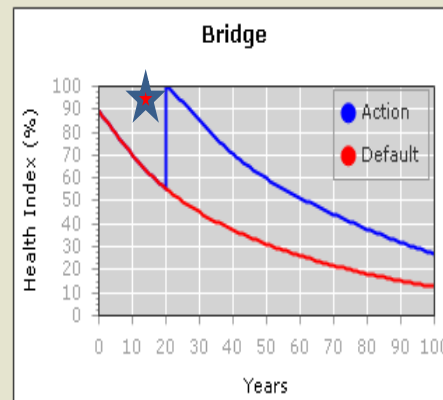
Effects on Each Utility Criterion

Category name	Before WC	After WC
Total Utility	49.44	50.5
Condition	57.16	59.73
Deck	5	5
Superstructure	81	81
Substructure	91	91
Scour	58	58
Element ratings	33.13	99.91
(12) Reinforced Concrete Deck	41.39	99.72
(331) Reinforced Concrete Bridge Railing	47.56	100
(510) Wearing Surfaces	10.44	100
Risk	53.81	53.81
Accident	53.81	53.81
Mobility	34.2	34.2
Percent of truck detoured.	34.2	34.2

Effects on Each Element

Element	Str. Unit	Env.	Condition	Effect
(12) Re Concrete Deck	0	Low (2)		
(331) Re Conc Bridge Railing	0	Low (2)		
(510) Wearing Surfaces	0			

Deterioration



Project Planning Preservation Actions

- Project Planning
 - Ability to create and view projects .
 - Define projects by grouping together work items and bridges
 - Determine cost and effectiveness of projects and the end result of performing the selected work on the selected bridges.
 - Dashboards to view higher level numbers and effects, while also being able to drill down to specific results and details

Advanced Inspections and 3D Visualization



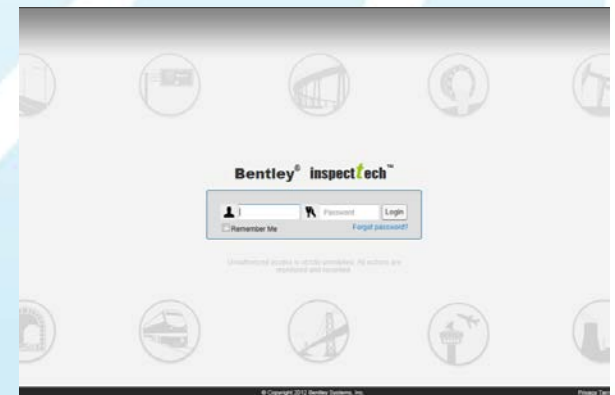
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InspectTech

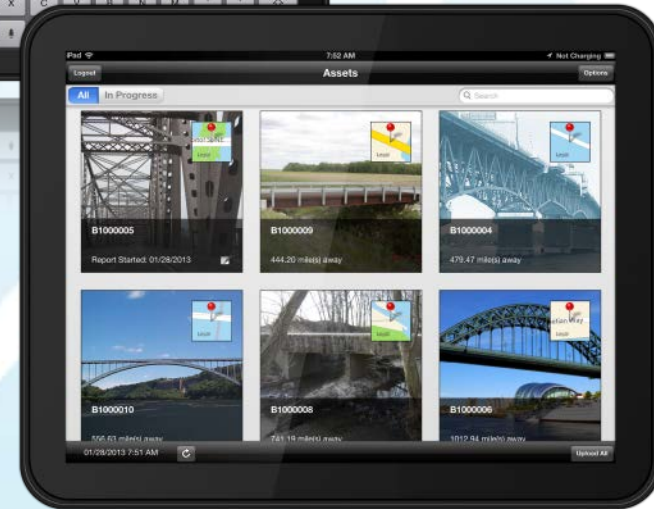
- Does inspections better, quicker, and more reliably
- Provides detailed work flows for field-to-office review and approval
- Supports various inspection types and agency-defined fields and forms
- Provides Web or field options
- Integrates inspector tools and reference material into one easy-to-use interactive form
- Licensed by 14 US DOTs and numerous transit, counties, cities



InspectTech Collector Mobile



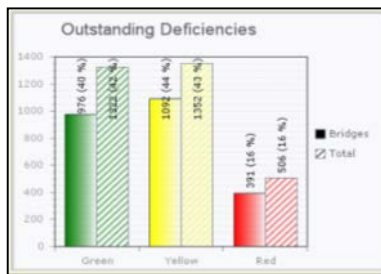
- Advanced mobility for users of **InspectTech**
- Sleek, graphical interface
- Custom inspection workflow
- Capture photos, video, audio
- Built-in historical asset information
- Calculations based on NBI
- Automatic report generation
- Cloud service synchronization with home system





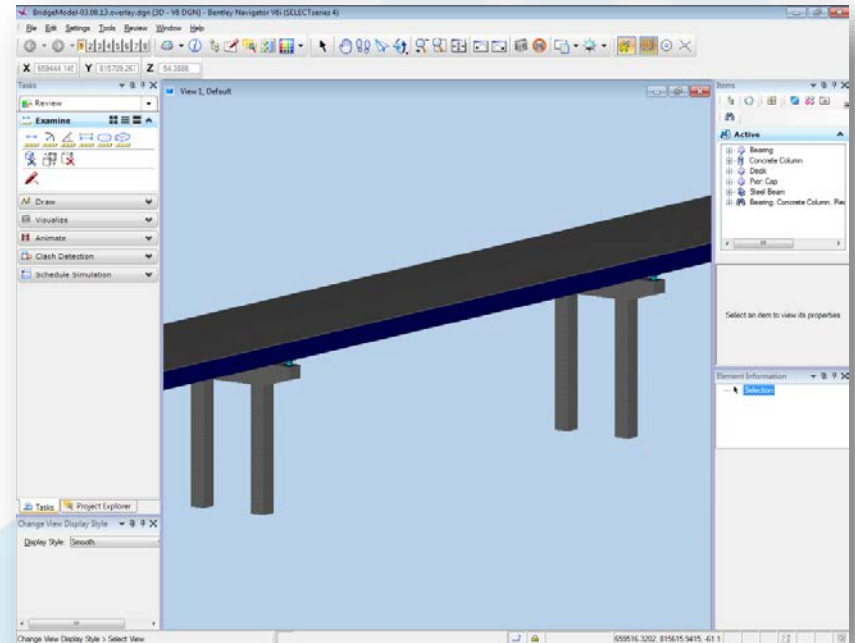
After extensive research Mn/DOT selected InspectTech software as the new Structure Information Management System

- Single site for instant entry/retrieval for over 16,000 structures and 500 users (state, local, consultant, and researchers)
- Secure SaaS solution available anytime
- Field module for remote entry
- Integrated maintenance system
- Significant time savings and error reductions during all phases
- Meets all federal regulations and guidelines
- Integrates with Pontis/BrM



The Design Model

- Physical Model
 - Analytical Model
- ↓
- CAD File
- ↓
- Published *iModel* / format



Physical Model in Viewer

- Viewed in Navigator or other program

Data to Operations - InspectTech Documents

The screenshot displays the Bentley Navigator V8i (SELECTseries 4) interface. The main window shows a 3D model of a bridge pier with a yellow circle highlighting a specific bearing component. The Project Explorer on the left lists the model structure, including 'Pier 23-24'. The Properties panel on the right provides detailed information about the selected 'Bearing - Friction Pendulum Bearing'.

Properties Panel Data:

Properties	
Catalog Type	Bearing
Catalog Instance	Friction Pendulum Bearing
Beam Line Number	5
GUID	9F2CBF0B-0B42-4180-9389-F
ID Asset Tag	
ID Description	
ID Item ID	
ID Keynote	
ID Name (Alternate)	
ID Notes	
ID Type ID	Westbound Bridge
Install Date	6/10/2013
Phasing	New Construction
Pier number	23
Product Manufacturer	Earthquake Protection System
Product Model	Single Pendulum Bearing
Structural Function	Structural
Structural Material	

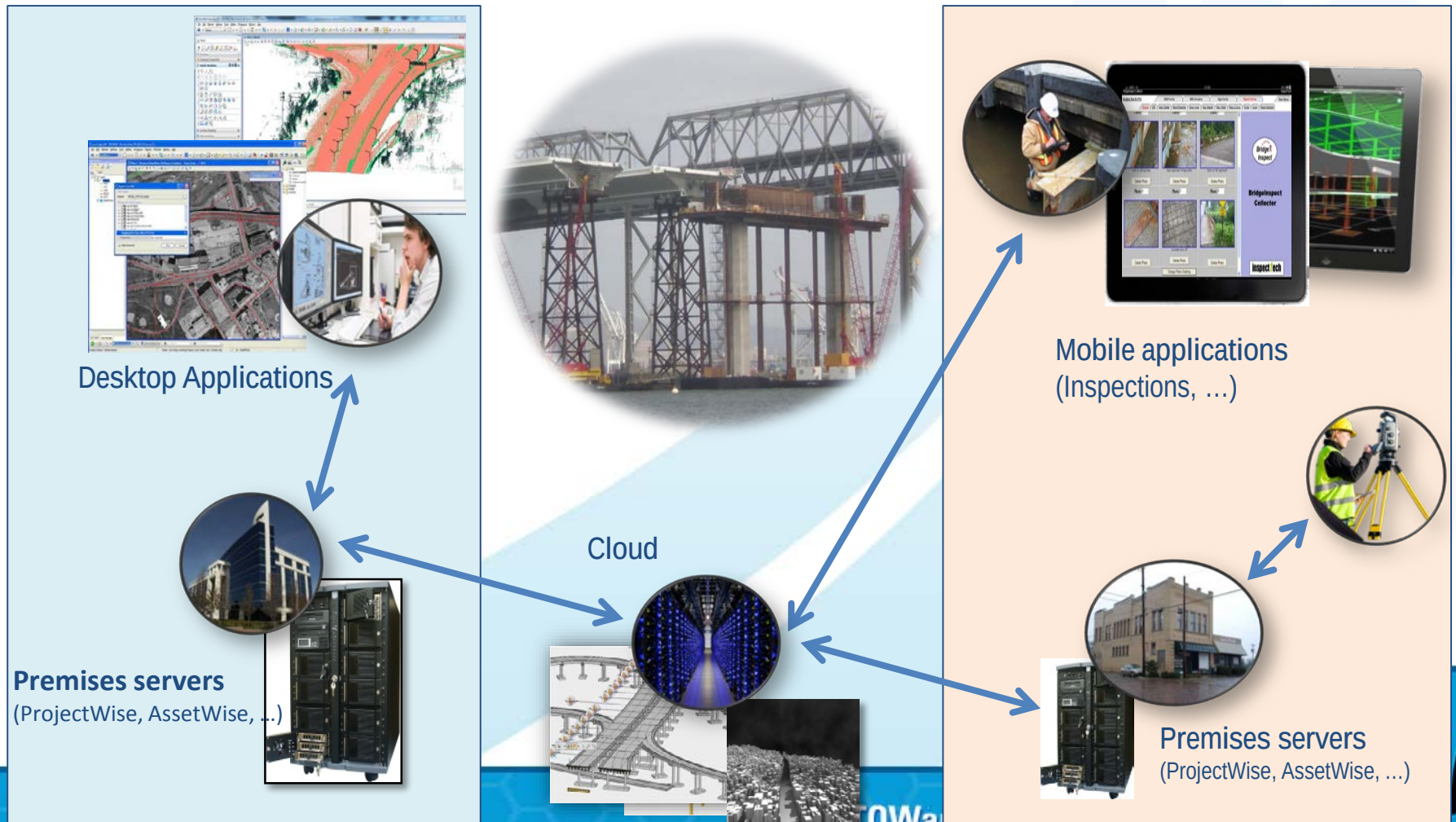
Links Panel Data:

Links	
Links	2
Links[0]	FP-Bearing.pdf
Links[1]	FP-Bearing_RM Bridge Ar

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Information Mobility



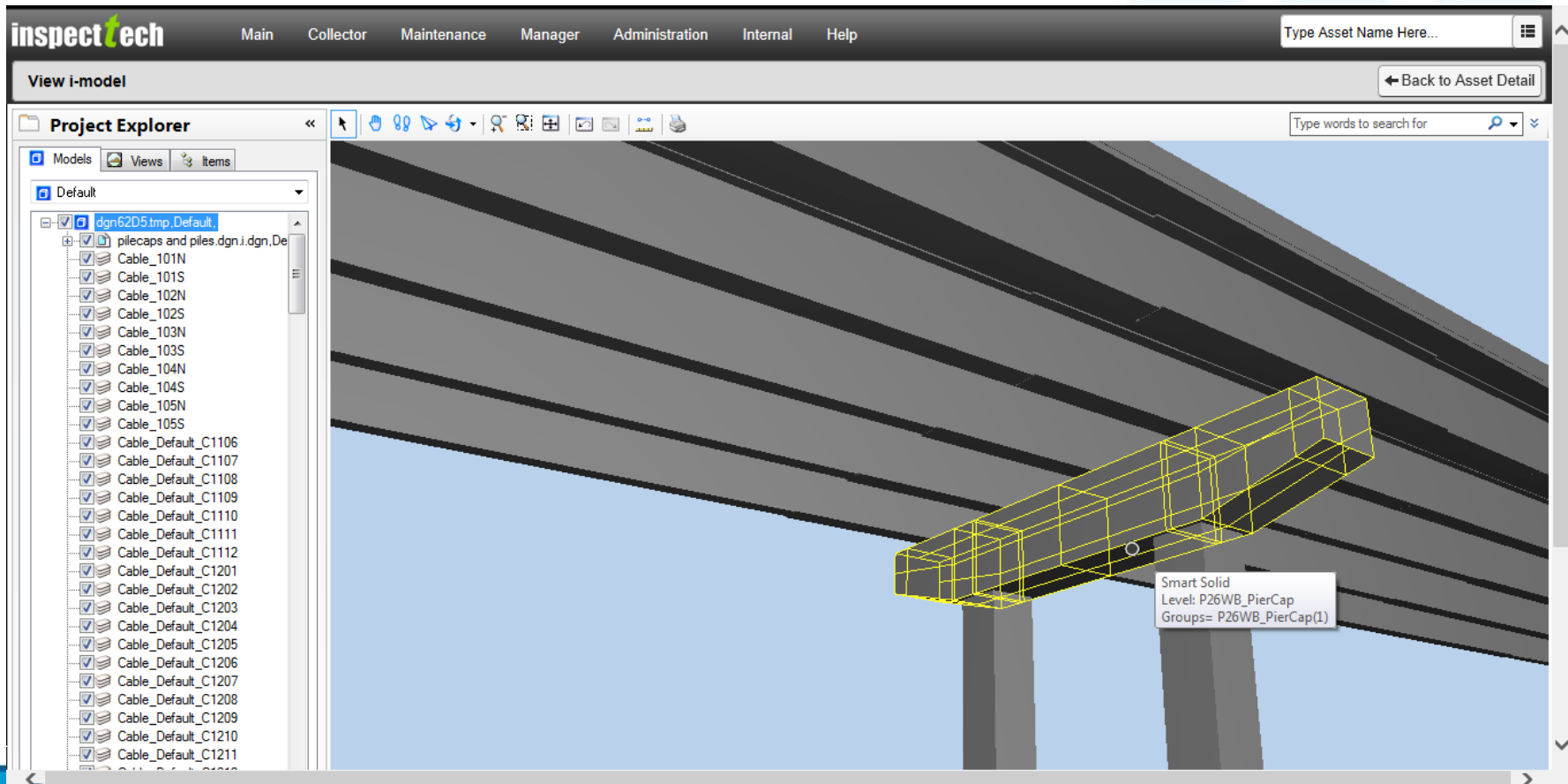
Goal: ~~100 Years~~ Indefinite Operation

- Preservation and detailed data is key
- 3D design/as-built model loaded directly into structure management system for BIM
- Component hierarchy and inventory data
- Data can be viewed and linked on interactive 3D model accessible via the Web



Detailed 3D views

- Collect and report condition data
- Plan maintenance actions with full history



Special Searching Criteria

- Search by entire structure, span or specific component type
- Ability to drill down into the specific parts of the asset tree to the desired level
- Can combine fields and criteria as needed to make simple or complex queries using boolean logic

Typical Approach (Large Bridge)

- Increased potential for typing errors from redundant entry of data
- Time-consuming task of manually retrieving data from multiple hard copies and databases
- Each biennial inspection report is ~2,000 pages
- Nearly impossible to try to manually do complex queries from paper reports
- In case of emergency – needed a better plan to obtain data
- Needed to centralize data to one location for easy access



интерпретация результатов

Inspection Field Notes

антисоциальная личность.

Inspection Field Notes

BrIM

Field Selections in Queries

The screenshot displays the AASHTOWare Bridge Management software interface. At the top, there are navigation tabs for 'Reports', 'Administration', and 'Help'. A 'Find an asset' link is visible in the top right corner. The main area shows a query configuration window titled 'List of values for: 58 - Deck (Rating)'. This window includes a 'Filter Values:' input field and a 'Filter' button. Below this is a table with a 'Value' column and a 'Select' column. The table lists 11 values, each with a corresponding 'Select' link. The values are: 0 - FAILED CONDITION (OUT OF SERVICE - BEYOND CORRECTIVE ACTION), 1 - "IMMINENT FAILURE CONDITION (BRIDGE CLOSED, PENDING CORRECTIVE ACTION), 2 - CRITICAL CONDITION (ADVANCE LOSS TO PRIMARY STRUCTURE, MAY CLOSE BRIDGE), 3 - SERIOUS CONDITION (PRIMARY STRUCTURE AFFECTED), 4 - POOR CONDITION (ADVANCED DETERIORATION), 5 - FAIR CONDITION (MINOR SECTION LOSS), 6 - SATISFACTORY CONDITION (MINOR DETERIORATION), 7 - GOOD CONDITION (SOME MINOR PROBLEMS), 8 - VERY GOOD CONDITION (NO PROBLEMS NOTED), 9 - EXCELLENT CONDITION, and N - Not Applicable. At the bottom of the window, there is a 'Run Query' button and a URL: /INDOT/bridgestat/loading_redirect.aspx?resolved=1&redirect=%2fINDOT%2fbridgestat%2fmodal_pages.

Reports Administration Help Find an asset

where field '58 - Deck (Rating)' = ''

☒ Add Criteria Field to Displayed Columns

Return results that match of the fol

58 - Deck (Rating) =

[Click to add a new criteria](#)

List of values for: 58 - Deck (Rating) Run Query

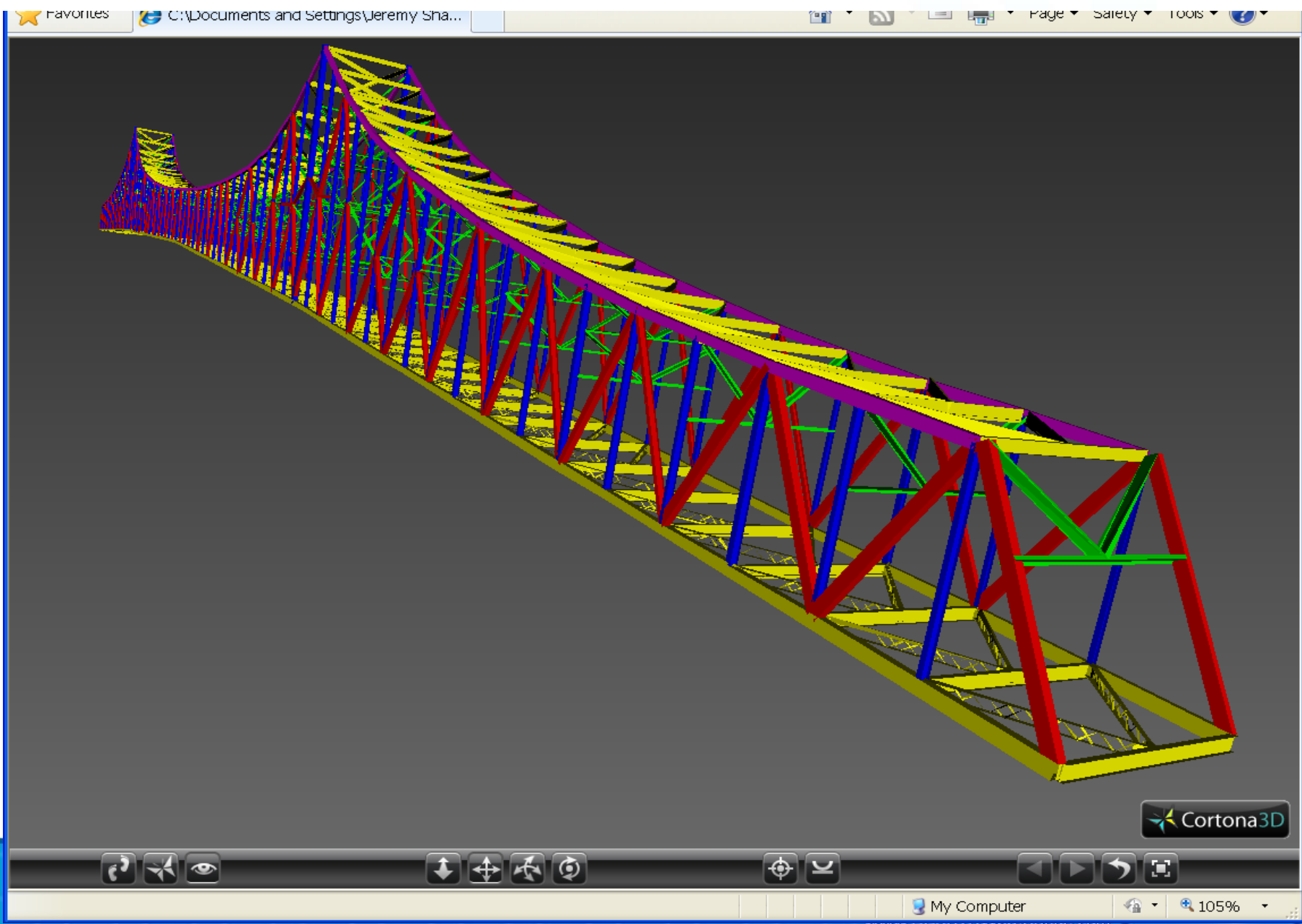
Filter Values: Filter

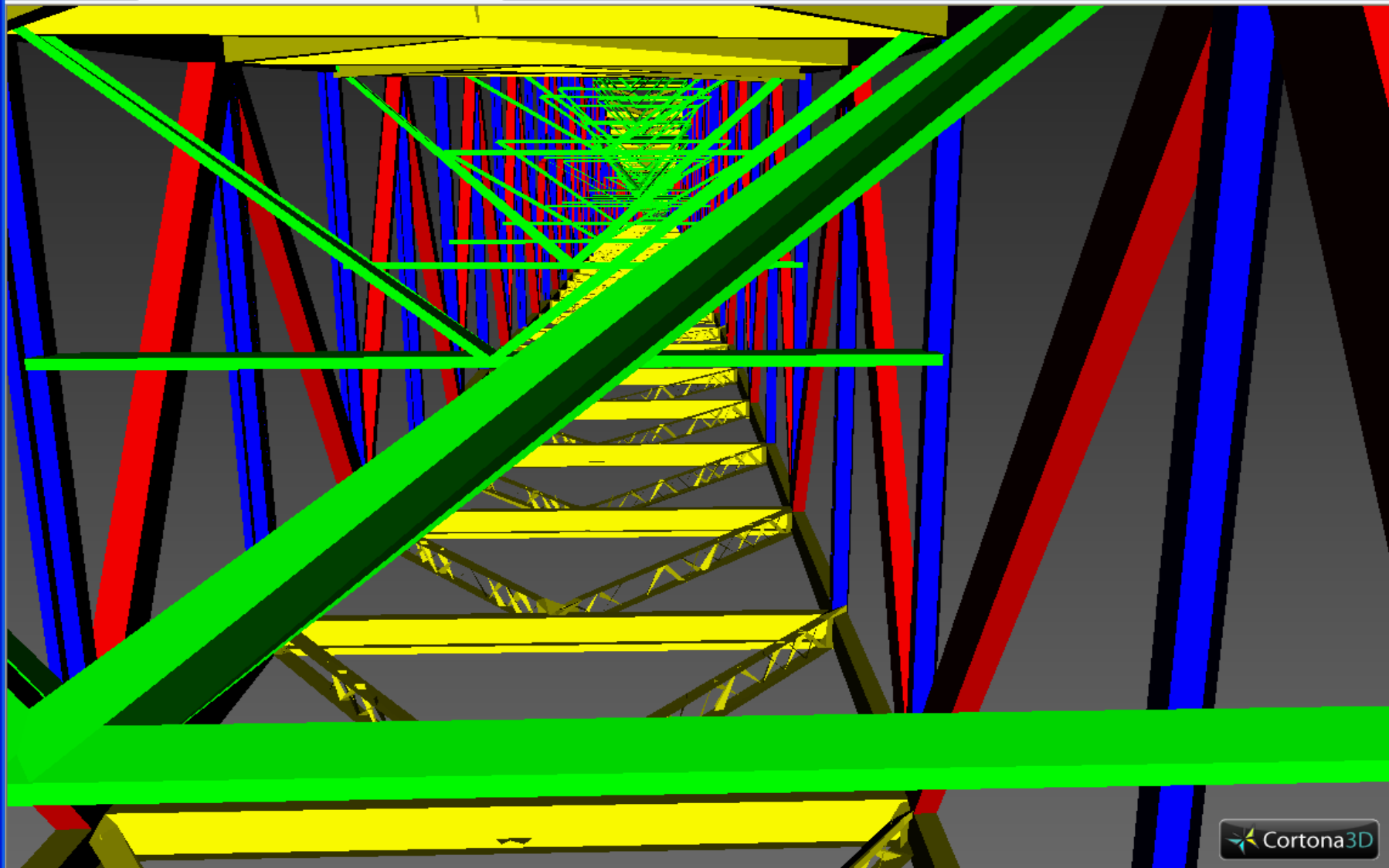
	Value
Select	0 - FAILED CONDITION (OUT OF SERVICE - BEYOND CORRECTIVE ACTION)
Select	1 - "IMMINENT FAILURE CONDITION (BRIDGE CLOSED, PENDING CORRECTIVE ACTION)
Select	2 - CRITICAL CONDITION (ADVANCE LOSS TO PRIMARY STRUCTURE, MAY CLOSE BRIDGE)
Select	3 - SERIOUS CONDITION (PRIMARY STRUCTURE AFFECTED)
Select	4 - POOR CONDITION (ADVANCED DETERIORATION)
Select	5 - FAIR CONDITION (MINOR SECTION LOSS)
Select	6 - SATISFACTORY CONDITION (MINOR DETERIORATION)
Select	7 - GOOD CONDITION (SOME MINOR PROBLEMS)
Select	8 - VERY GOOD CONDITION (NO PROBLEMS NOTED)
Select	9 - EXCELLENT CONDITION
Select	N - Not Applicable

11 values found

/INDOT/bridgestat/loading_redirect.aspx?resolved=1&redirect=%2fINDOT%2fbridgestat%2fmodal_pages.

Run Query





Cortona3D

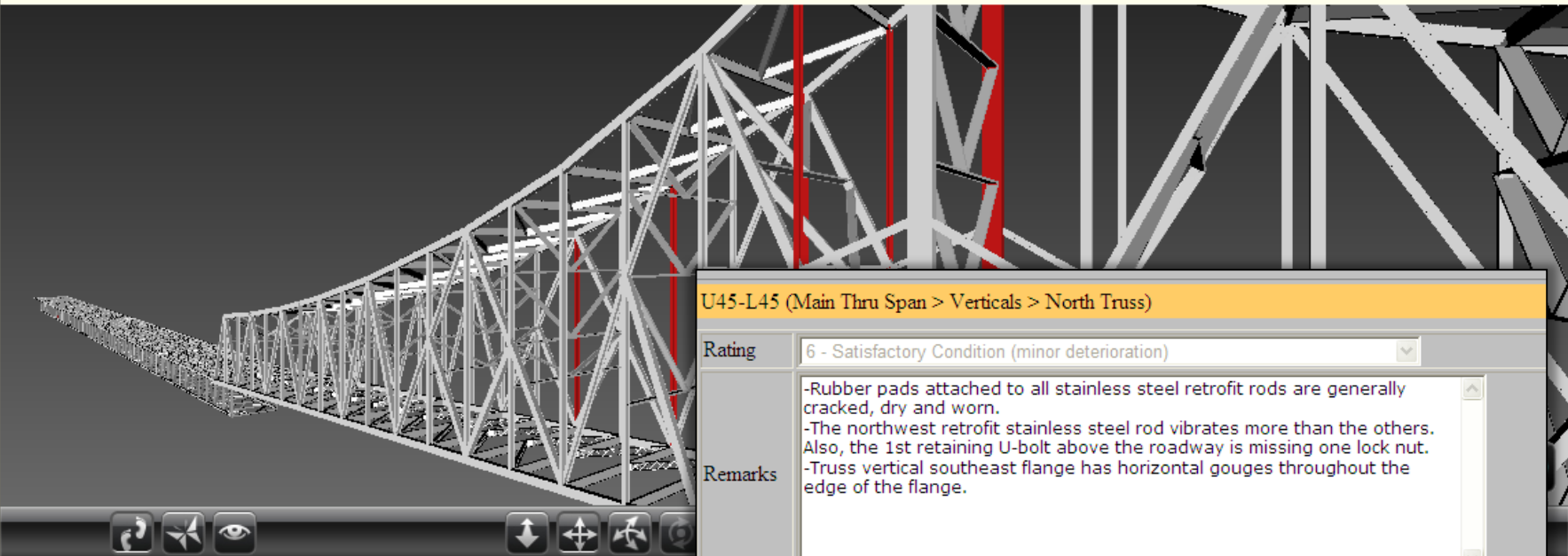
Preservation Support – Usable Data

StructureSuite

inspect**tech**

Quick Select: [View asset](#)

[Main](#) [GIS](#) [Query](#) [Reports](#) [Administration](#) [Help](#)



U45-L45 (Main Thru Span > Verticals > North Truss)

Rating: 6 - Satisfactory Condition (minor deterioration)

Remarks:

- Rubber pads attached to all stainless steel retrofit rods are generally cracked, dry and worn.
- The northwest retrofit stainless steel rod vibrates more than the others. Also, the 1st retaining U-bolt above the roadway is missing one lock nut.
- Truss vertical southeast flange has horizontal gouges throughout the edge of the flange.

Photo: 64

Photo: 65

Photo: 66

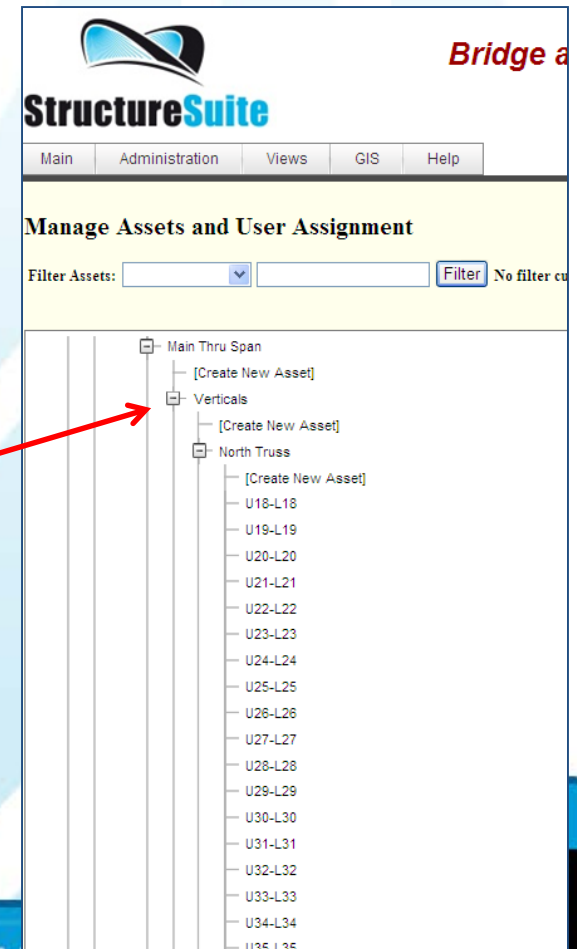
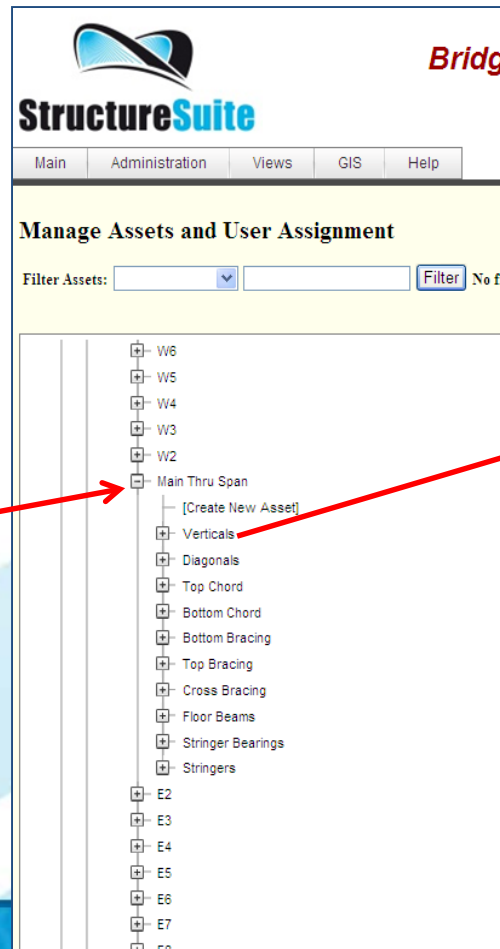
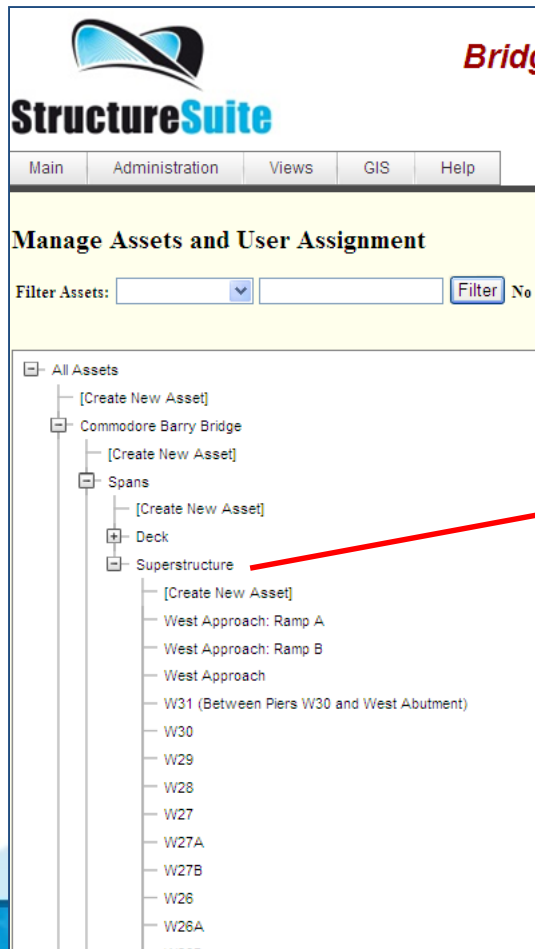
Conditions of Use

Details

- 3D Solid Model
- Represent only the details that user cares about
- Utilize color for different layers – condense to single color for search results
- Ability to turn on/off layers
- Information all driven off database and web-interface

Drill down into detail of bridge to add ratings, comments, pictures

User drilling down to bridge, span, superstructure, main span, verticals, north truss and specific parts

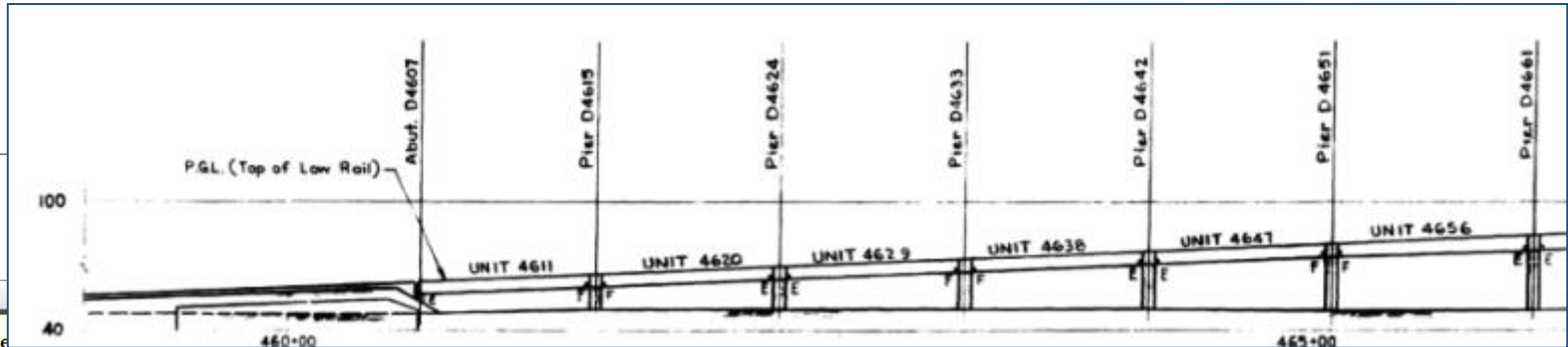


DC Metro – 2D Approach



Main Administration

Minnesota Avenue Aerial



Inspection Date	Asset Code	Asset Name	Owner	Asset Type	Submitted To	Status	
12/08/2009	Minnesota Avenue Aerial Structure	Minnesota Avenue Aerial Structure	Webb, Gerald	Bridge		Approved on 6/24/2010	✖

Minnesota Avenue Aerial Structure Sub-Assets:

Show Assets in

by

Displayed Inspection Type:

Inspection Date	Asset Code	Asset Name	Sub-Assets	Owner	Asset Type	Submitted To	Status	
12/08/2009	1116600	Abutment 2432 IB&OB	None	Webb, Gerald	Bridge		Approved on 6/24/2010	✖
12/08/2009	1116700	Span 2437 IB	None	Webb, Gerald	Bridge		Approved on 6/24/2010	✖
12/03/2009	1116800	Span 2437 OB	None	Limones, Jose	Bridge		Approved on 6/24/2010	✖
12/03/2009	1116900	Crossbox/Pier 2442 IB&OB	None	Limones, Jose	Bridge		Approved on 6/24/2010	✖
12/08/2009	1117000	Span 2447 IB	None	Webb, Gerald	Bridge		Approved on 6/24/2010	✖
12/03/2009	1117100	Span 2447 OB	None	Limones, Jose	Bridge		Approved on 6/24/2010	✖
12/03/2009	1117200	Crossbox/Pier 2452 IB&OB	None	Limones, Jose	Bridge		Approved on 6/24/2010	✖
12/08/2009	1117300	Span 2457 IB	None	Webb, Gerald	Bridge		Approved on 6/24/2010	✖
12/03/2009	1117400	Span 2457 OB	None	Limones, Jose	Bridge		Approved on 6/24/2010	✖
02/04/2010	1117500	Crossbox/Pier 2462 IB&OB	None	Webb, Gerald	Bridge		Approved on 6/24/2010	✖
02/04/2010	1117600	Span 2467 IB	None	Webb, Gerald	Bridge		Approved on 6/24/2010	✖
12/07/2009	1117700	Span 2467 OB	None	Limones, Jose	Bridge		Approved on 6/24/2010	✖



AASHTOWare Bridge Management

www.AASHTOWareBridge.com

Conclusions

- Software solutions are rapidly developing the analytical engines to quantifying and describe benefits of preservation actions
- Need data on benefits of each action
 - What is the benefit of bridge washing?
- Goal is to provide easy to use, understandable tools for practitioners

Questions?

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